

KAUSTUBH DATTATRAY SHINDE

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PROFESSIONAL SUMMARY

Electronics & Telecommunication Engineer with hands-on full-stack development experience across Java/Spring Boot, React and SQL, alongside applied AI and embedded-systems project work. Built and shipped a complete Spring Boot + React e-commerce platform end to end, and an AI-powered autonomous vehicle combining computer vision, NLP and real-time control. Comfortable working across the stack, debugging distributed systems, and quickly picking up new frameworks and tools to solve real problems.

EDUCATION

ISBM College of Engineering, Pune

2022 – 2026

Bachelor of Engineering — Electronics & Telecommunication Engineering, Savitribai Phule Pune University (SPPU)

CGPA: FY 7.59 | SY 9.11 | TY 8.80 | BE

RESEARCH PUBLICATIONS

Advanced Safety System for Two-Wheeler Using PIC16F877A and CAN Bus

ICETT-2026

International Conference on Emerging Trends in Technology — Co-author

- Co-authored a research paper proposing a low-cost embedded safety system for two-wheelers built around a PIC16F877A microcontroller with CAN bus communication for reliable, fault-tolerant data exchange.
- Contributed to the system architecture covering impact detection, CO gas monitoring and PT100-based engine temperature sensing, with GSM-based emergency alerting.

AI-Powered Autonomous RC Vehicle: Design and Performance

Final Year Project Report

SPPU Stage-I Report — Author

- Authored the Phase-I project report covering literature survey, problem formulation, system architecture and proposed methodology for an AI-powered autonomous RC vehicle.
- Detailed the integration of voice-command recognition, conversational AI processing, motor control and computer-vision-based obstacle detection into a single Raspberry Pi platform.

PROJECTS

ShopEase | Full-Stack E-Commerce Platform

Java, Spring Boot, Spring Data JPA, MySQL, React, Vite, Maven

- Architected a full-stack e-commerce platform with a Spring Boot + Spring Data JPA + MySQL backend and a React (Vite, React Router) frontend, spanning product catalog, cart, checkout, order tracking and an admin dashboard.
- Designed the relational data model using JPA entities (Product, Order, OrderItem, ProductReview) with one-to-many and element-collection mappings for product variants, images, specifications and reviews.
- Built REST APIs for product retrieval, order creation, order tracking by unique order number, and admin order-status updates, each wired to a dedicated React page.
- Implemented transactional order-confirmation and status-update emails via Spring Mail, and a persistent cart on the frontend using local storage with a data-mapper pattern to sanitize product data.

AI-Powered Autonomous RC Car | Voice Control & Navigation

Raspberry Pi, Python, Flask, OpenCV, Gemini API, GPIO

- Built an autonomous RC vehicle on a Raspberry Pi 4 with a Flask backend, live OpenCV video streaming, and a Gemini vision-model agent that analyses the camera feed to make real-time driving decisions.
- Implemented a multithreaded GPIO motor-control module for the drive system, with an independent safety loop to manage obstacle response and emergency stop.
- Developed a voice-command pipeline that routes transcribed natural-language instructions through a Gemini-powered agent to executable drive commands (forward, backward, left, right, stop).
- Built a browser-based control dashboard serving a live MJPEG camera stream alongside manual drive controls and an autopilot toggle.

Amazon Clone | E-Commerce UI

JavaScript (ES6 Modules), HTML5, CSS3, Jasmine

- Built a responsive Amazon-style storefront in vanilla JavaScript, HTML and CSS with a cart, multi-step checkout, delivery-option selection and an order-tracking view, persisting cart state via local storage.
- Refactored the cart module into a class-based (OOP) design and wrote unit tests for cart and currency-formatting utilities using the Jasmine testing framework.

Responsive YouTube Clone | Front-End Layout

HTML5, CSS3 (Flexbox, Grid)

- Built a fully responsive YouTube-style interface using semantic HTML5 and modern CSS, including a collapsible sidebar, search header and video-grid layout.
- Implemented adaptive, multi-breakpoint layouts using CSS Flexbox and Grid.

React Chatbot | Conversational UI

React, JavaScript, Vite

- Developed a chatbot interface in React using functional components and hooks to manage conversation state and render message history.
- Structured the UI into reusable ChatInput, ChatMessage and ChatMessages components, bundled and built with Vite.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL, HTML5, CSS3

Frameworks: React, Spring Boot, Spring Data JPA

Database: MySQL

Developer Tools: Git, GitHub, Docker, Linux, VS Code, Raspberry Pi, Docker Hub

Concepts: Object-Oriented Programming, Data Structures & Algorithms, REST APIs, Responsive Web Design

Certifications: Java Programming Masterclass (Udemy), React Developer (HackerRank), SQL Advanced (HackerRank)